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22MAT230 MATHEMATICS FOR COMPUTING 4

An FFT-Based Audio Watermarking Method with Optimized Embedding

First Review report

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1 Abstract

Audio watermarking is an effective technique for embedding imperceptible information into audio signals to support copyright protection, authentication, and ownership verification. Unlike image and video watermarking, audio watermarking must carefully account for the sensitivity of the human auditory system while maintaining robustness against common signal processing operations. This project presents the design and implementation of a frequency-domain audio watermarking system based on frame-wise processing and the Fast Fourier Transform (FFT). A watermark sequence is embedded into selected frequency components of the host audio signal in a controlled manner to achieve a balance between imperceptibility and reliable detection. The embedded watermark is detected using a correlation-based detection approach that determines the presence of the watermark without requiring access to the original audio signal. The system is implemented and evaluated using MATLAB, and its performance is analyzed in terms of audio quality and watermark detectability. The results demonstrate that frequency-domain audio watermarking provides an effective framework for embedding copyright information while preserving perceptual audio quality.

2 Introduction

Audio watermarking is a signal processing technique used to embed imperceptible information into an audio signal for purposes such as copyright protection, ownership verification, authentication, and broadcast monitoring. With the rapid growth of digital media distribution and online streaming platforms, protecting audio content from unauthorized copying and misuse has become increasingly important. Unlike cryptographic methods, which secure content during transmission, watermarking provides persistent protection by embedding information directly into the signal itself.

Audio watermarking presents unique challenges when compared to image and video watermarking. The human auditory system is highly sensitive to distortions, even at very low levels, which places strict constraints on how much a signal can be modified without introducing perceptible artifacts. In addition, audio signals are often subjected to common signal processing operations such as compression, filtering, resampling, and noise addition, all of which can degrade or destroy an embedded watermark if it is not designed carefully.

To address these challenges, watermarking techniques are broadly classified into time-domain and frequency-domain methods. Time-domain approaches typically modify audio samples directly and are computationally simple, but they often suffer from poor robustness against signal processing attacks. Frequency-domain techniques, on the other hand, embed watermark information into transformed representations of the signal, such as the Fast Fourier Transform (FFT) domain, which allows better control over perceptual impact and improved resistance to common distortions.

This project focuses on the design and implementation of a frequency-domain audio watermarking system based on frame-wise processing and FFT analysis. By operating on short-time frames of the audio signal, the method exploits the quasi-stationary nature of audio over small intervals and enables selective modification of frequency components. A correlation-based detection strategy is employed to determine the presence of the embedded watermark without requiring access to

the original audio signal. The system is implemented using MATLAB and serves as a practical framework for studying the trade-offs between imperceptibility and reliable watermark detection in audio signals.

3 Literature Review

Table 1: Summary of Existing Audio Watermarking Techniques

Ref.	**Method / Domain**	**Key Idea**	**Limitations**
[1]	Robust Audio Watermarking in the Time Domain	Embeds watermark directly by modifying audio samples in the time domain with low computational complexity.	Poor robustness against signal processing attacks such as filtering, resampling, and compression.
[2]	Muteness-Based Audio Watermarking Technique	Embeds watermark information in silent or low-energy regions to achieve high imperceptibility.	Limited robustness and low embedding capacity; watermark is vulnerable to noise and cropping.
[3]	Audio Watermarking: Features, Applications and Algorithms	Provides a comprehensive survey of watermarking domains, applications, and trade-offs between imperceptibility, robustness, and capacity.	Does not propose a concrete embedding algorithm; primarily descriptive in nature.
[4]	Robust Audio Watermarking based on HWD and SVD	Uses transform-domain decomposition and singular value modification to improve robustness.	High computational complexity and reduced suitability for real-time implementation.
[5]	FFT-Based Audio Watermarking (Base Paper)	Embeds watermark into prominent peaks of the FFT magnitude spectrum of non-overlapping frames, enabling blind detection using correlation.	Uses fixed embedding parameters and does not explicitly optimize the trade-off between imperceptibility and detection reliability.

Several audio watermarking techniques have been reported in the literature, spanning time-domain, frequency-domain, and hybrid approaches. Time-domain methods offer simplicity but suffer from poor robustness against common signal processing operations [1]. Muteness-based techniques improve imperceptibility by exploiting low-energy regions, though they are vulnerable to noise and cropping [2]. Survey studies provide a broad classification of watermarking algorithms and their trade-offs [3], while transform-domain approaches using wavelets and singular value decomposition achieve robustness at the cost of increased computational complexity [4]. The FFT-based watermarking scheme described in [5] forms the foundation for this project and motivates further analysis and implementation.

4 Objectives

- To study the fundamental concepts of digital audio watermarking and their role in copyright protection.
- To analyze existing audio watermarking techniques and identify their limitations in terms of imperceptibility and robustness.
- To design a frequency-domain audio watermarking scheme based on frame-wise processing and the Fast Fourier Transform (FFT).
- To implement the proposed audio watermarking system using MATLAB.
- To detect the embedded watermark using a correlation-based detection approach without requiring the original audio signal.
- To evaluate the performance of the watermarking system in terms of audio quality and watermark detectability.

5 Proposed Methodology

5.1 System Overview

The proposed audio watermarking system embeds imperceptible watermark information into an audio signal in the frequency domain while enabling reliable blind detection. The system follows a frame-based processing approach using the Fast Fourier Transform (FFT). Watermark embedding is performed by modifying selected frequency-domain magnitude components of each frame, and detection is achieved using a correlation-based decision rule without requiring access to the original audio signal.

5.2 Audio Signal Representation

Let $x[n]$ denote the discrete-time audio signal, where $n = 0, 1, \ldots, N-1$. The signal is assumed to be sampled at a fixed sampling rate and normalized to a suitable amplitude range. Since audio signals are non-stationary in nature, the signal is divided into short-time frames over which it can be approximated as stationary.

5.3 Frame Segmentation

The audio signal is divided into K non-overlapping frames, each of length L samples. The k-th frame is given by

$$x_k[n] = x[kL + n], \quad n = 0, 1, \ldots, L-1, \tag{1}$$

where $k = 0, 1, \ldots, K-1$. Frame-wise processing allows the watermark to be distributed across the entire audio signal.

5.4 Frequency-Domain Transformation

Each frame $x_k[n]$ is transformed into the frequency domain using the FFT:

$$X_k[m] = \sum_{n=0}^{L-1} x_k[n] e^{-j2\pi mn/L}, \quad m = 0, 1, \ldots, L-1. \tag{2}$$

The complex FFT coefficients are expressed as

$$X_k[m] = A_k[m] e^{j\phi_k[m]}, \tag{3}$$

where $A_k[m]$ and $\phi_k[m]$ denote the magnitude and phase spectra, respectively.

5.5 Watermark Sequence Generation

The watermark is represented as a bipolar sequence

$$\mathbf{w} = \{w_1, w_2, \ldots, w_P\}, \quad w_i \in \{-1, +1\}, \tag{4}$$

where P denotes the watermark length. The sequence is designed to have zero mean and controlled energy to facilitate correlation-based detection.

5.6 Key-Based Selection of Embedding Frequency Bins

Watermark embedding is performed by selecting frequency bins from a predefined candidate set $\mathcal{C}$, typically chosen from a mid-frequency range to balance imperceptibility and robustness. To enhance security, the selection of embedding bins is performed in a key-dependent pseudo-random manner.

Let $\mathcal{C} = \{c_1, c_2, \ldots, c_{M_{\max}}\}$ denote the candidate frequency bin set. Using a secret key K, a pseudo-random number generator is seeded to select P distinct indices from $\mathcal{C}$. The resulting embedding set is given by

$$\mathcal{M} = \mathrm{sort}\big(\mathrm{randperm}_K(\mathcal{C}, P)\big), \tag{5}$$

where $\mathrm{randperm}_K(\cdot)$ denotes key-dependent random permutation. Since the same key is used during detection, the embedding locations can be reproduced reliably.

5.7 Watermark Embedding Formulation

For each selected frequency bin $m_i \in \mathcal{M}$, watermark embedding is performed by modifying the FFT magnitude as

$$A'_k(m_i) = A_k(m_i) + \alpha w_i, \tag{6}$$

where α is the embedding strength parameter. The modified FFT coefficient is then reconstructed as

$$X'_k(m_i) = A'_k(m_i) e^{j\phi_k(m_i)}. \tag{7}$$

All non-selected frequency bins remain unchanged.

5.8 ADMM-Based Optimization for Watermark Embedding

To control the trade-off between imperceptibility and watermark detectability, the embedding process is formulated as an optimization problem. Let $\mathbf{a} \in \mathbb{R}^P$ denote the vector of original magnitudes at the selected bins and $\mathbf{d}$ denote the magnitude modification vector such that

$$\mathbf{a}' = \mathbf{a} + \mathbf{d}. \tag{8}$$

The detection statistic is defined as

$$D = \mathbf{a}'^T \mathbf{w}, \tag{9}$$

and reliable detection requires

$$(\mathbf{a} + \mathbf{d})^T \mathbf{w} \geq T_{\min}. \tag{10}$$

The optimization problem is therefore

$$\min_{\mathbf{d}} \frac{1}{2} \|\mathbf{d}\|_2^2 \quad \text{subject to} \quad (\mathbf{a} + \mathbf{d})^T \mathbf{w} \geq T_{\min}. \tag{11}$$

This problem is solved using the Alternating Direction Method of Multipliers (ADMM).

5.9 Reconstruction of the Watermarked Audio Signal

After computing the optimized modification vector $\mathbf{d}$, the modified FFT coefficients are reconstructed using the original phase information. The inverse FFT is applied to obtain the watermarked frame:

$$x'_k[n] = \text{IFFT}\{X'_k[m]\}. \tag{12}$$

The complete watermarked audio signal is obtained by concatenating all reconstructed frames in their original temporal order.

5.10 Watermark Detection and Decision Rule

At the receiver, the received audio signal is processed using the same framing, FFT, and key-based bin selection. The magnitudes of the selected bins form the detection vector $\mathbf{z}$. A correlation-based decision statistic is computed as

$$D = \sum_{i=1}^{P} z_i w_i. \tag{13}$$

The watermark is declared present if

$$D \geq T, \tag{14}$$

and absent otherwise. This enables blind watermark detection without requiring the original host signal.

6 Results and Discussion

6.1 Overview of Experimental Evaluation

The performance of the proposed audio watermarking system was evaluated using a MATLAB-based implementation. The experiments were designed to assess the correctness of watermark embedding, imperceptibility of the watermarked audio, reliability of blind watermark detection, and robustness under common signal processing attacks. A binary watermark of length 100 was embedded using key-based frequency bin selection and ADMM-based optimization. Detection was performed without access to the original audio signal. Figure 1,Figure 2,Figure 3 respectively show the representation of audio in time and its magnitude along with the generated watermark sequence.

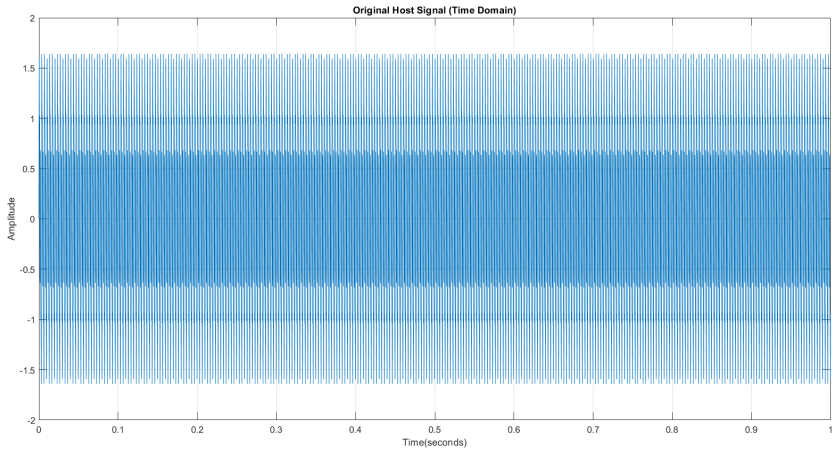

Figure 1: Host Audio signal in time domain

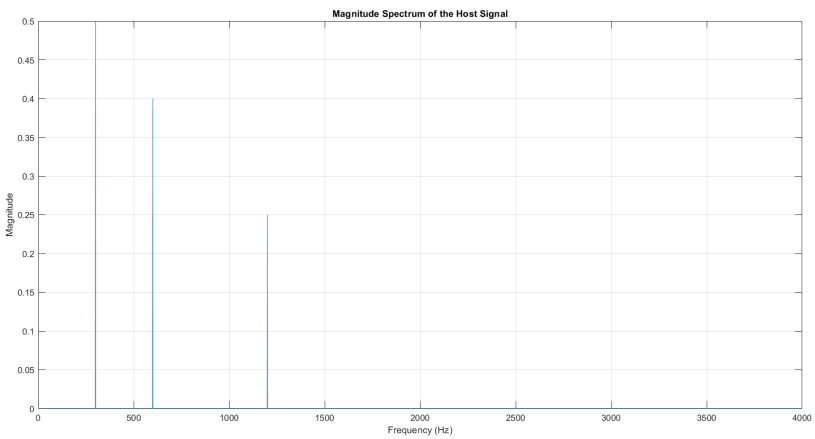

Figure 2: Magnitude of Host Audio signal

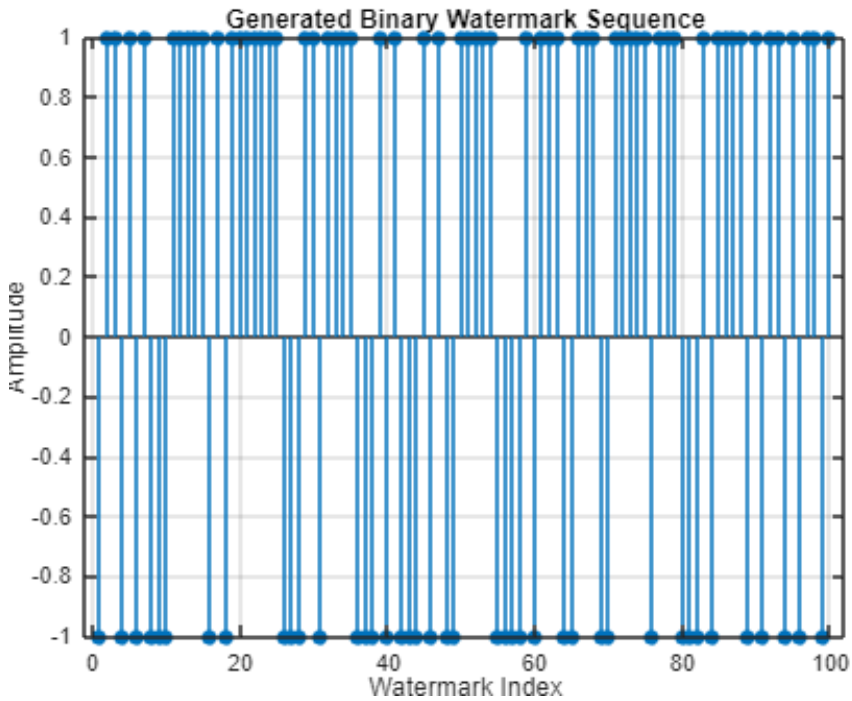

Figure 3: Generated Watermark sequence

6.2 Key-Based Frequency Bin Selection Analysis

Figure 4 illustrates the magnitude spectrum of the host audio signal along with the selected frequency bins used for watermark embedding. The continuous curve represents the host audio spectrum, while the stem markers indicate the selected bins. The embedding locations are sparsely distributed within a mid-frequency range and are selected in a key-dependent pseudo-random manner. This approach enhances security by ensuring that the embedding locations are reproducible only with the correct key, while also minimizing perceptual distortion by avoiding perceptually sensitive low-frequency components.

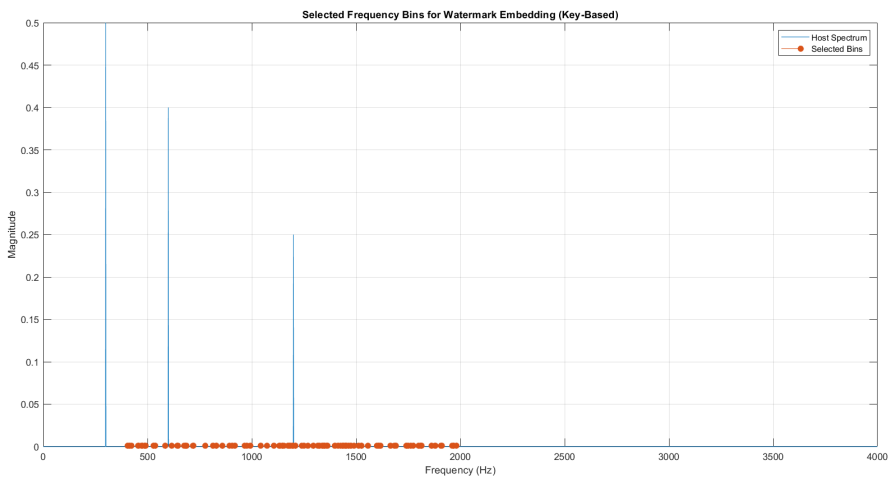

Figure 4: Host audio spectrum with key-based selected frequency bins for watermark embedding.

6.3 ADMM Convergence and Constraint Satisfaction

The ADMM-based optimization plays a central role in controlling the trade-off between imperceptibility and watermark detectability. After convergence, the correlation value $\mathbf{w}^T\mathbf{d}$ was observed to be equal to 1, satisfying the imposed detection constraint. The difference between successive iterations was found to be 6.4136×10^{-5}, indicating numerical convergence. These results confirm that the optimization process successfully enforces the detection constraint while introducing minimal distortion to the host signal.

6.4 Detection Threshold and Decision Logic

Watermark detection is performed using a correlation-based decision rule. The detection score D is compared against a predefined threshold T to determine the presence or absence of the watermark. For the original (non-watermarked) audio signal, the detection score D_0 was approximately -8.99×10^{-13}, which lies well below the threshold. In contrast, the clean watermarked signal produced a detection score of 0.01, which exceeds the threshold. This clear separation ensures reliable blind detection with negligible false positives.

6.5 Detection Results for Clean and Original Audio

Table 2 and Figure 5 summarizes the detection results for the original and clean watermarked audio signals. The results demonstrate a clear distinction between watermark-absent and watermark-present cases, validating the correctness of the embedding and detection framework.

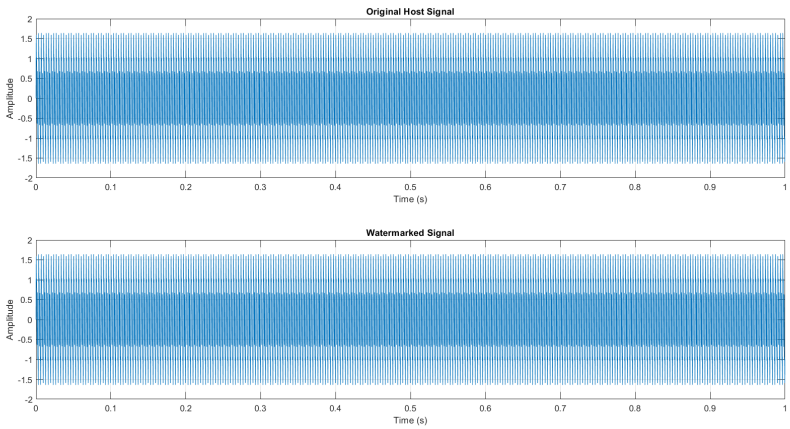

Figure 5: Original Signal V/s Watermarked Signal.

Table 2: Detection results for original and clean watermarked audio

Signal Type	Detection Score
Original (No watermark)	-8.99×10^{-13}
Clean Watermarked	0.01

6.6 Imperceptibility Analysis

Imperceptibility was evaluated using the signal-to-noise ratio (SNR) between the original and watermarked audio signals. An SNR value of 94.8058 dB was obtained, indicating that the watermark embedding introduces negligible distortion. Such a high SNR confirms that the watermark is perceptually inaudible, which can be attributed to frequency-domain magnitude modification, phase preservation during reconstruction, and distortion minimization enforced by the ADMM optimization.

6.7 Robustness Analysis Under Signal Processing Attacks

The robustness of the proposed watermarking scheme was evaluated under several common signal processing attacks. The detection scores obtained under different attack scenarios are summarized in Table 3 and illustrated in Figure 6.

6.7.1 No Attack (Clean Watermarked Signal)

For the clean watermarked signal, the detection score was 0.01, and the watermark was correctly detected. This serves as the baseline performance of the system.

6.7.2 Additive White Gaussian Noise (AWGN)

Under AWGN attack, the detection score dropped to -0.18437, resulting in watermark detection failure. This behavior is expected, as additive noise significantly distorts the frequency-domain magnitude components used for correlation-based detection.

6.7.3 Low-Pass Filtering

When low-pass filtering was applied, the detection score was reduced to -0.010239, and the watermark was not detected. This occurs because the filtering operation removes or attenuates the frequency bins used for watermark embedding.

6.7.4 Amplitude Scaling

For amplitude scaling, the detection score was 0.008, and the watermark was successfully detected. This result indicates that the correlation-based detector is largely invariant to amplitude scaling, demonstrating robustness against gain changes.

Table 3: Detection scores under different attack conditions

Attack Type	**Detection Score**
Original (No watermark)	-8.99×10^{-13}
Clean Watermarked	0.01
AWGN Attack	-0.18437
Low-Pass Filtering	-0.010239
Amplitude Scaling	0.008

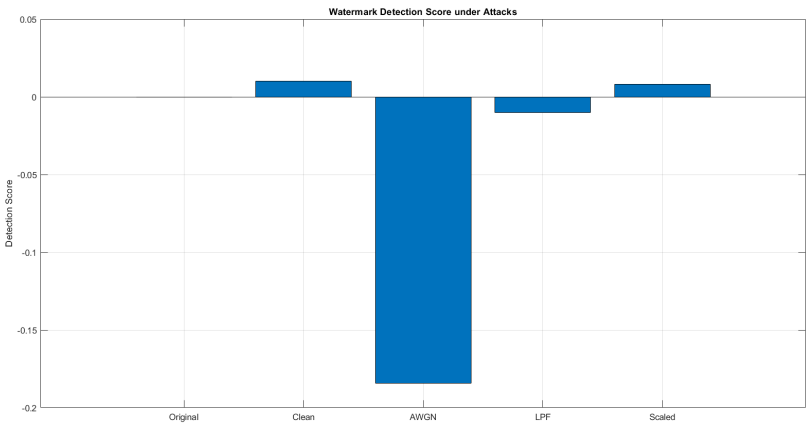

Figure 6: Detection scores under clean and attacked conditions.

6.8 Comparative Interpretation and Limitations

The experimental results demonstrate that the proposed system achieves reliable blind detection and high imperceptibility under clean conditions. The ADMM-based embedding strategy provides controlled distortion while ensuring detection reliability. However, the system exhibits limited robustness against additive noise and filtering attacks, which is a known challenge for frequency-domain watermarking schemes without redundancy or synchronization mechanisms. These limitations highlight opportunities for further enhancement.

6.9 Summary of Results

In summary, the proposed FFT-based audio watermarking system successfully embeds imperceptible watermark information and enables reliable blind detection under clean conditions. The system demonstrates high audio quality and robustness to amplitude scaling, while its sensitivity to noise and filtering motivates future improvements aimed at enhancing robustness.

7 Future Works(for final review)

- Evaluate the robustness of the proposed watermarking system under common signal processing attacks such as noise addition, filtering, resampling, compression, and cropping.
- Perform a detailed quantitative analysis using additional performance metrics such as bit error rate (BER) and receiver operating characteristic (ROC) curves.
- Compare the performance of the proposed method with existing watermarking techniques, including the base paper, under identical experimental conditions.
- Investigate adaptive embedding strategies where the embedding strength and threshold parameters are dynamically adjusted based on signal characteristics.

- Extend the current implementation to support multi-bit watermark embedding and improved synchronization mechanisms.

8 Conclusion

This project focused on the design and implementation of a frequency-domain audio watermarking system for copyright protection. By employing frame-wise processing and Fast Fourier Transform (FFT) analysis, watermark information was embedded into selected frequency components while preserving perceptual audio quality. An optimization-based embedding strategy using the Alternating Direction Method of Multipliers (ADMM) was incorporated to control distortion and ensure reliable watermark detection. The complete system was implemented in MATLAB, providing a practical framework for further evaluation and enhancement of audio watermarking techniques.

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